

Decentralized Free-Support Wasserstein Barycenter

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Introduction

Wasserstein barycenters average probability measures while preserving the geometry of mass. In decentralized settings, each node sees only a local measure and communicates with neighbors. Existing decentralized WB methods usually fix a common support grid, reducing the task to consensus over weights (Dvurechensky et al., NeurIPS 2018; Kroshnin et al., ICML 2019). Free support removes that common coordinate system: atom locations move, atom identities become ambiguous, and the optimization is nonconvex.

Theory

Theorem: exact aggregation descends

In a fully connected network,

$$F(\mathbf{B}^{(r+1)}) \leq F(\mathbf{B}^{(r)}) - \eta(2 - \eta)\Delta^{(r)}.$$

Hence

$$\min_{r < R} \Delta^{(r)} \leq \frac{F(\mathbf{B}^{(0)}) - F_m^{\inf}}{R\eta(2 - \eta)}.$$

The interpolation step is a relaxed MM update with an $\mathcal{O}(1/R)$ stationarity rate.

Corollary: Sinkhorn variant descends

The entropy-regularized objective obeys the same descent form,

$$F_\tau(\mathbf{B}^{(r+1)}) \leq F_\tau(\mathbf{B}^{(r)}) - \eta(2 - \eta)\Delta_\tau^{(r)}.$$

Regularized local OT keeps the same convergence logic.

Theorem: topology adds a floor

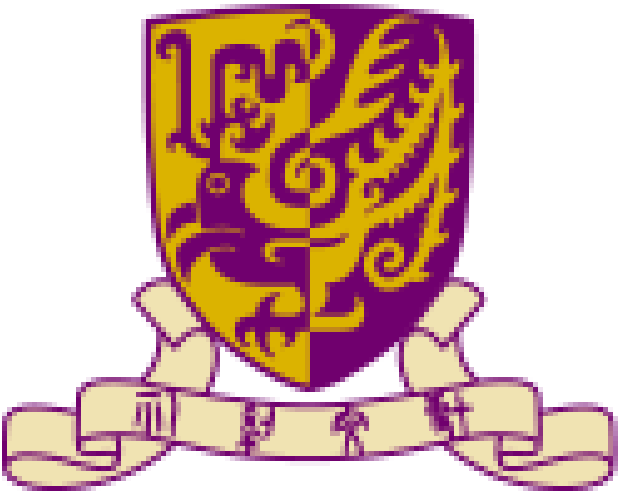
Over arbitrary connected graphs,

$$\min_{r < R} \Delta_\tau^{(r)} \leq \mathcal{O}(1/R) + C \sum_{s < R} \delta_s^2 / R.$$

Gossip mixing gives $\delta_{r+1} \leq \rho^L(L_\tau \delta_r + \zeta_\tau)$.

Finite-step gossip turns exact descent into inexact descent controlled by consensus error.

For more methodology and theoretical details, please refer to the full paper.



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Decentralized WB can learn free supports by aligning moving atom trajectories, not fixed grids.



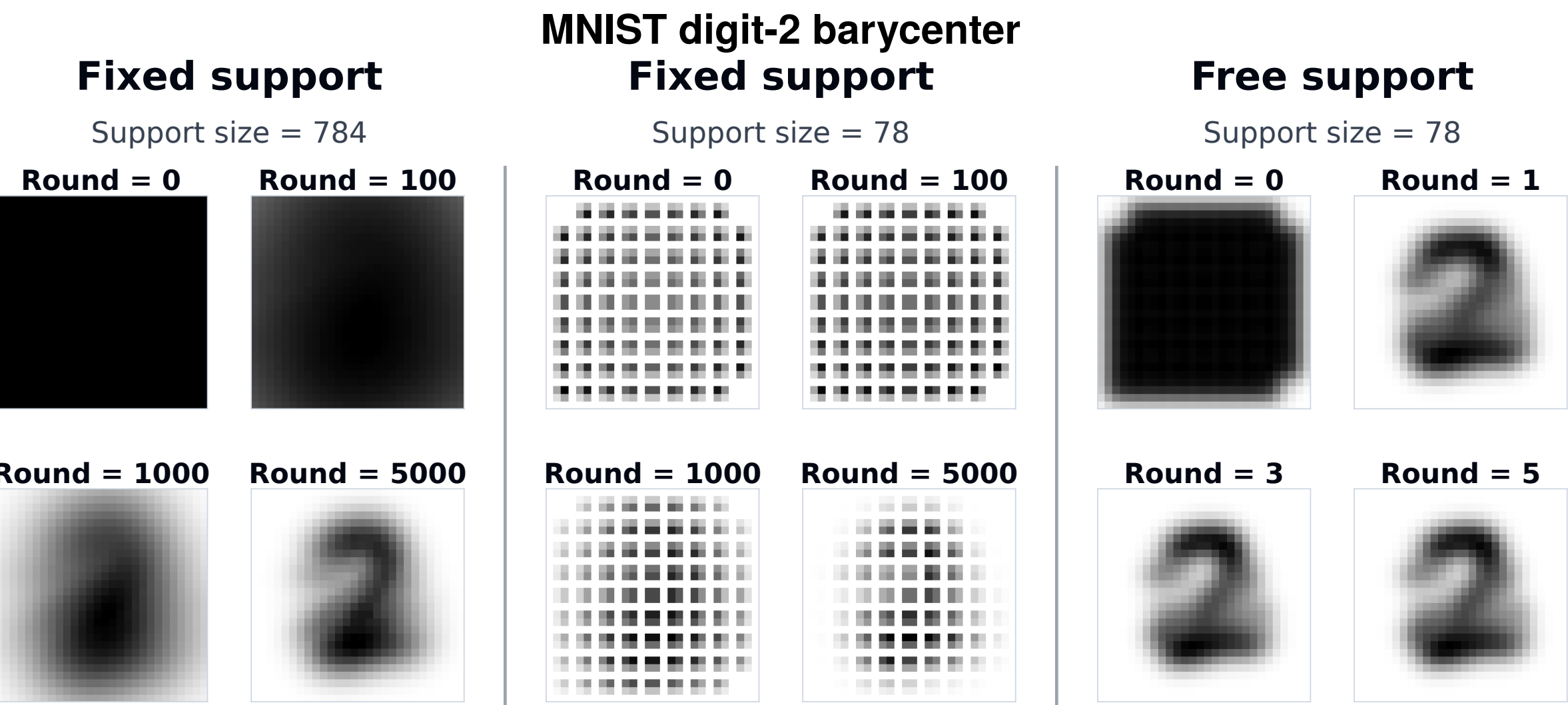
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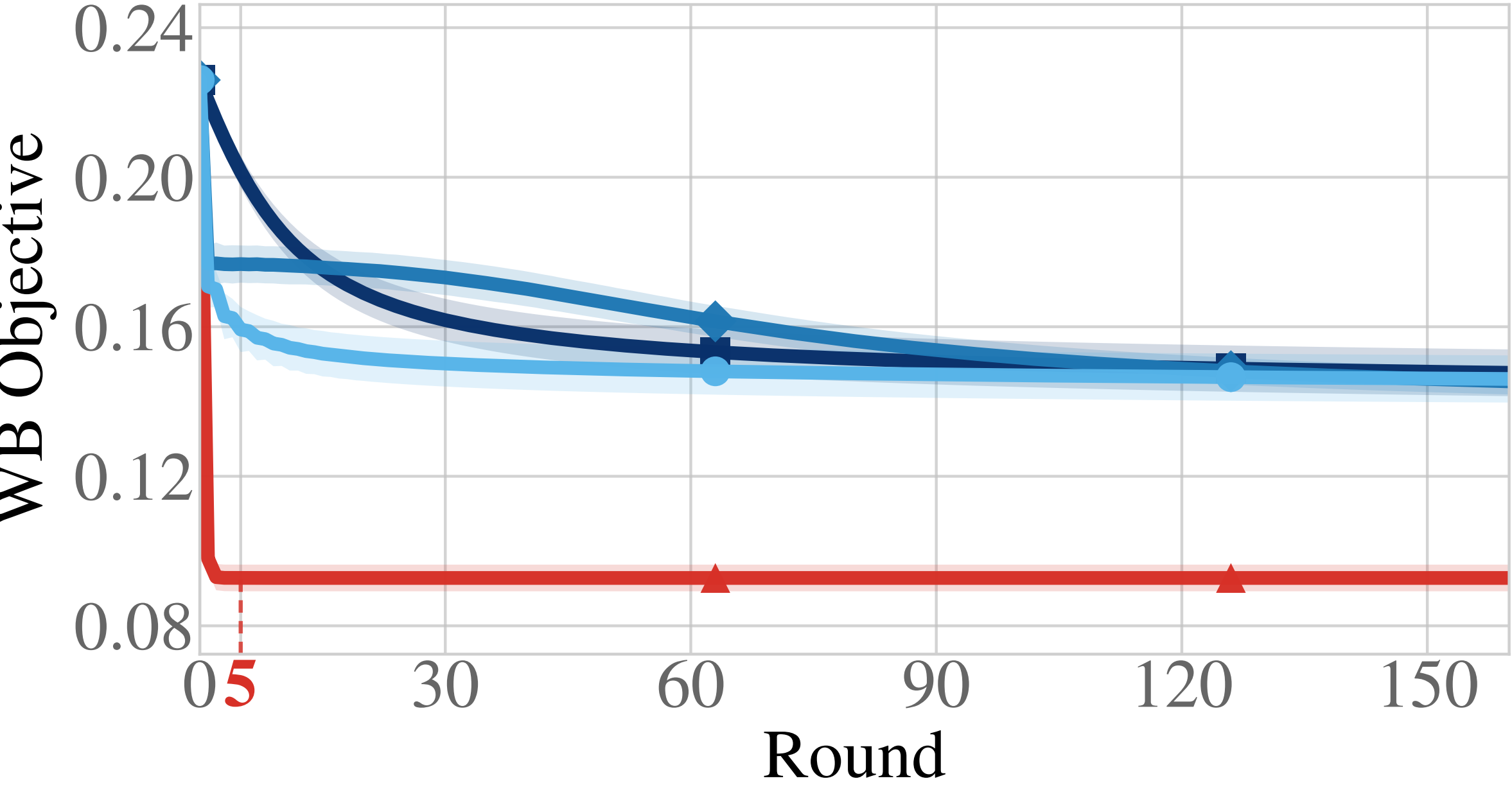
Method

Each agent computes an OT plan from its current barycenter estimate to its local measure, moves atoms along McCann interpolation, and then gossips only the $d \times m$ support matrix. Atom identities are inherited along continuous support trajectories, making averaging of moving supports well-defined.

Primary Evidence



Free support reaches lower evaluated WB objective after 5 rounds while the fixed-support baseline uses 5000 rounds on the 784-pixel grid.



Experimental Takeaway

- MNIST: objective drops from 0.00825 to 0.00682, a 17.34% reduction.
- Resource use falls to 0.00995% compute and 0.0199% communication.
- Truncated Gaussians: objective 0.0886 versus 0.1371–0.1392 for fixed-support decentralized baselines.

References

Agueh and Carlier, SIAM J. Math. Anal. 2011; Cuturi and Doucet, ICML 2014; Dvurechenskii et al., NeurIPS 2018; Kroshnin et al., ICML 2019.
For additional experimental results, please refer to the full paper.

